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Product Mapping Schema

Branch

feature/product-mapping

Objective

The Product Mapping schema is designed to support AI-powered product compatibility checks, avatar-product matching, recommendation systems, and virtual try-on functionality.

Schema Fields

Field	Type	Description
productId	ObjectId	References the associated product
category	String	Product category (Men, Women, Kids, etc.)
clothType	String	Type of clothing (Shirt, T-Shirt, Dress, Jeans, etc.)
supportedBodyTypes	Array[String]	Body types supported by the product
compatibility	String	Compatibility status for AI processing

avatarCompatibility	Array[String]	Supported avatar types
createdAt	Date	Record creation timestamp

Compatibility Workflow

User Avatar



Backend Checks Compatibility



Matching Products Selected



AI Try-On Starts

Workflow Explanation

1. User uploads or selects an avatar.
2. Backend retrieves avatar attributes.
3. ProductMapping collection is queried.
4. Compatible products are filtered based on:
 - category
 - cloth type
 - body type
 - avatar compatibility
5. Matching products are returned.
6. AI Try-On service generates the preview.

AI Integration Purpose

The Product Mapping schema provides structured compatibility data required by AI services.

Features Supported

- Avatar-to-product matching
- AI virtual try-on validation
- Product filtering
- Compatibility scoring
- Personalized recommendations

Benefits

- Improved try-on accuracy
- Faster compatibility checks
- Better recommendation quality
- Reduced AI processing errors

Recommendation Engine Usage

The schema can be used by recommendation services to:

- Suggest products based on body type
- Recommend compatible clothing categories
- Filter unsupported products
- Personalize shopping experiences

Example Use Case

If a user has an athletic body type and uses a male avatar, the system can retrieve products that support:

`supportedBodyTypes = Athletic`

`avatarCompatibility = Male Avatar`

Testing Performed

Backend Startup Test

Command:

```
npm run dev
```

Expected Result:

```
MongoDB Connected
```

```
Server Running Successfully
```

Validation Test

Verified:

- ProductMapping schema loads successfully
- No schema compilation errors
- No startup failures
- MongoDB connection remains functional

Deliverables

- ProductMapping.js created
- Schema fields implemented
- Compatibility workflow documented
- AI integration purpose documented
- Recommendation engine support documented
- Backend tested successfully